



Module 2

NFS Version 3

About This Module



This module focuses on enabling you to do the following:

- Describe the features of NFS Version 3 (NFSv3)
- Describe the implementation of NFSv3 on NetApp storage
- Describe NFS exports
- Explain NFSv3 security using ONTAP software



Lesson 1

Overview of NFSv3

NFSv3 Features

- Provides safe, asynchronous writes:
 - NFSv3 enables response to a client even when a write to disk has not occurred.
 - NFSv3 works well with FAS NVRAM.
 - The server can buffer multiple write requests and issue a single, more efficient write request.
 - ONTAP software responds with a COMMIT response when a file is written.
 - The NFS write call can request a STABLE (SYNC) or UNSTABLE (ASYNC) write.
 - ONTAP software always responds as STABLE when queried:

```
# tcpdump -nv | grep 2049
...
client.3...>system.2049:200 write fh x... 23 (23) bytes @ 0<unstable>
system.2049>client.3...reply ok 160 write POST:REG 644 ids 0/0 sz 23 23 bytes
```

- Supports 64-bit file sizes and offsets
- Enables client access to files larger than 2GB

NFSv3 Capabilities

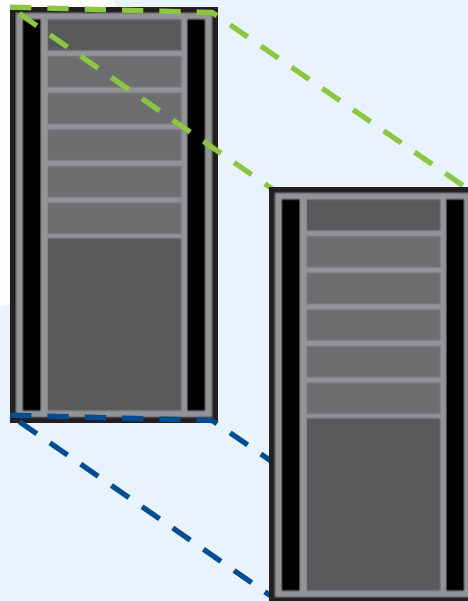
- Close-to-open file consistency:
 - Compares GETATTR operation of previously cached files to the GETATTR operation, when the file is opened next
 - Uses a data cache (if the same results occur) or purges the cache (if results are not the same)
 - Uses `-cto` to enable or `-nocto` to disable (enabled by default)
- Dynamic retry:
 - Enables the client to retry previous NFS calls with no response by server
 - Can be configured by the client
- READDIRPLUS enhancements:
 - Returns file handles and attributes in an extended directory list
 - Makes it unnecessary for the client to request the server for each entry

ONTAP 9 and NFS Client Support

- NFSv3 is enabled by default in ONTAP 9 software and later.
- What clients can do:
 - Support NFSv3, NFS4, and NFS 4.1
 - Issue a mount:
 - `# mount -t nfs -o nfsvers=3 svmNFS:/ /svmNFS`
- Typical client mount options:
 - `rsize` and `wsiz` specify the buffer size for read and write requests (which are autonegotiated between the NFS client and server but can be modified):
ONTAP software sets the default `rsize` and `wsiz` to 65536 bytes.
 - `hard` specifies if the program accessing a file via NFS should stop and wait for the server to come back online, if the server is unavailable.
 - `intr` allows requests to be interrupted if the server cannot be reached.
 - `bg` forces the mount attempts to be run in the background (initial).

Use Cases

- NFSv3 is the version that is deployed most often.
- NFSv3 is used in the following scenarios:
 - File server in Linux- or UNIX-based environments
 - Virtualization environments such as VMware NFS datastore
 - Environments in which security is not a significant concern (private networks)
 - Certain database environments, such as Oracle



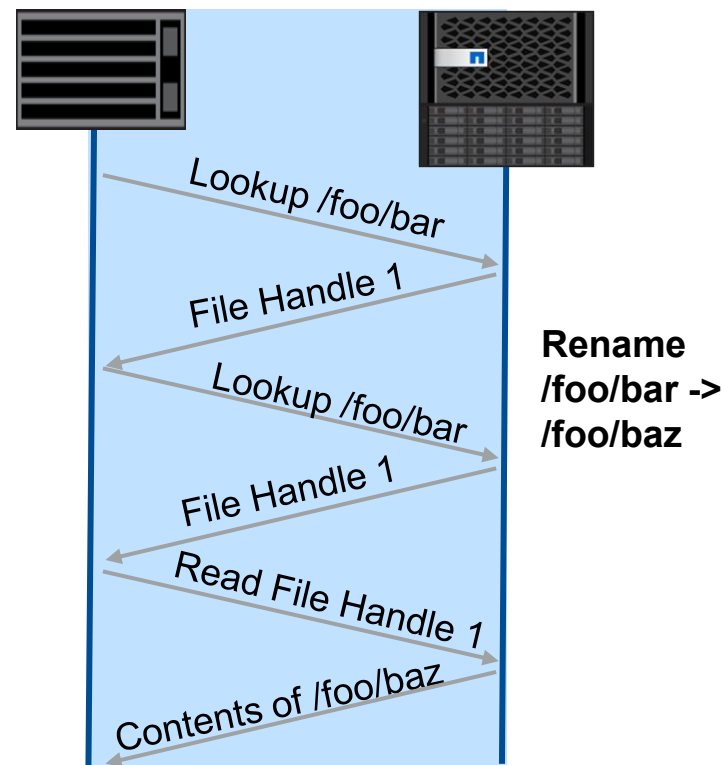


Lesson 2

NFSv3 Implementation

NFSv3 Client-Server Communication

- Characteristics of file handles:
 - A basic unit of NFS file access
 - A client reference for a file on the server
 - Created on the server
 - Used by the client
- Properties:
 - Isolation between client and server:
 - Preserves access through client disconnect and reconnect
 - Increases availability
 - Path name independence:
 - Survives renames
 - Increases flexibility



NFSv3 Example

NFSv3 Implementation

- Configure an NFS license NFS.
- Create or identify a storage virtual machine (SVM).
- Verify whether NFS access is enabled.

```
cluster1::>vserver create
    -vserver svmNFS
    -aggregate aggr1_cluster1_01
    -rootvolume svmNFS_root
    -rootvolume-security-style unix
cluster1::>vserver add-protocols
    -protocols SMB,fcp,iscsi
    -vserver svmNFS
```

```
cluster1::>vserver nfs show
```

```
Vserver:   svmNFS
```

General Access:	true
v3:	enabled
v4.0:	disabled
4.1:	disabled
UDP:	disabled
TCP:	disabled
Default Windows User:	-
Default Windows User:	-

NFSv3 Implementation

Network configuration (subnet method)

- Create or use an Ipspace:

```
cluster1::>network ipspace create -ipspace ipsp1
```

- Create or use a broadcast domain:

```
cluster1::>network port broadcast-domain create -broadcast-domain  
bcast1 -mtu 1500 -ports cluster1-01:e0e, cluster1-02:e0e
```

- Create or use a subnet:

```
cluster1::>network subnet create -subnet-name sub1  
-broadcast-domain bcast1 -ipspace ipsp1 -subnet  
192.168.0.0/24 -ip-range 192.168.0.101-192.168.0.150
```

- Create or use a logical interface (LIF):

```
cluster1::>network interface create -vserver svml -lif data1  
-data-protocol nfs -subnet sub1
```

NFSv3 Implementation

Enabling NFS

ONTAP CLI

```
cluster1::>vserver nfs create  
  -vserver svmNFS  
  -access true  
  -v3 enabled  
  -udp false  
  -tcp true
```

Default read/write sizes of
65,536 bytes

NetApp recommends mounting with TCP.

NFSv3 Implementation

Hostname: DNS and local file configuration

ONTAP CLI

```
cluster1::>vserver services dns create
      -vserver svmNFS
      -domains example.com
      -name-servers 192.168.239.3
```

```
cluster1::>vserver services dns hosts create
      -vserver svmNFS
      -address 192.168.239.4
      -hostname server.example.com
      -aliases test
```

NFSv3 Implementation

Netgroup: Local file configuration

ONTAP CLI

Upload an edited netgroup definition file.

```
cluster1::>vserver services name-service netgroup load  
-vserver svmNFS  
-source http://server.example.com/corp-netgroup  
cluster1::*>vserver services netgroup status
```

Verify that netgroups have replicated through the cluster.

NFSv3 Implementation

Network Information System (NIS) configuration

ONTAP CLI

```
cluster1::>vserver services nis-domain create
    -vserver svmNFS
    -domain nisdomain
    -active true
    -server server.example.com | -server 192.0.2.180
```

- You can create multiple NIS domains, but only one domain can be active at a time.
- No cache (slave) support exists in ONTAP 9 software.

NFSv3 Implementation for Active Directory

LDAP schema definitions

ONTAP CLI

```
cluster1::>vserver services ldap client create
  -client-config MY_LDAP
  -servers 10.10.10.100
  -ad-domain domain.netapp.com
  -bind-as-SMB-server true
  -schema AD-IDMU
  -port 389
  -query-timeout 3
  -min-bind-level sasl
  -base-dn DC=domain,DC=netapp,DC=com
  -base-scope subtree
  -preferred-ad-servers 10.10.10.10
```

Example of an Active Directory environment

NFSv3 Implementation for non-Active Directory

LDAP schema definitions

ONTAP CLI

```
cluster1::>vserver services ldap client create
  -client-config MY_LDAP
  -schema RFC-2307
  -servers 10.10.10.101
  -port 389
  -query-timeout 3
  -min-bind-level simple
  -base-dn DC=openldap,DC=netapp,DC=com
  -base-scope subtree
```

Example of a non-Active Directory LDAP (such as OpenLDAP)

NFSv3 Implementation

Lightweight Directory Access Protocol (LDAP) configuration

ONTAP CLI

```
Cluster1::>vserver services ldap show
```

Vserver	Client Configuration	Client Enabled
-----	-----	-----
svmNFS	MY_LDAP	true

NFSv3 Implementation

Username: File configuration

ONTAP CLI

```
cluster1::>vserver services name-service unix-user create
      -vserver svmNFS
      -user student2
      -id 501
      -primary-gid 500
```

```
cluster1::>vserver services name-services unix-user show
```

Vserver	User Name	User ID	Group ID	Full Name
-----	-----	-----	-----	-----
svmNFS	nobody	65535	65535	
svmNFS	pcuser	65534	65534	
svmNFS	root	0	1	
svmNFS	student2	501	500	



Lesson 3

NFS Exports

NFSv3 Export Policy Configuration

- The export policy specifies access permission to volumes to one or more clients that are specified by a host name, IP address, network mask, and netgroup.
- The default policy starts with no rules and enables no access, unless you use the NetApp ONTAP System Manager wizard to create the policy.
- The `-policy` attribute is used to associate the export policy with the volume:
 - This attribute is initially set to the `default` policy.
 - The `default` policy has no rules; you must create rules to configure it.
 - Set the `-policy` attribute to blank to disable this feature.
- Optionally, use `cmd vserver export policy` creates a policy with new rules.

NFSv3 Implementation

Creating an export policy and rule

Use additional commands to show, modify, rename, or delete the export policy.

ONTAP CLI

```
cluster1::>vserver export-policy create
    -vserver svmNFS
    -policyname read_only_expolicy
cluster1::>vserver export-policy rule create
    -vserver svmNFS
    -policyname read_only_expolicy
    -ruleindex 1
    -protocol any
    -clientmatch .example.com
    -rorule any
    -rwrule "krb5,sys"
    -anon 0
    -allow-suid false
    -allow-dev false
```

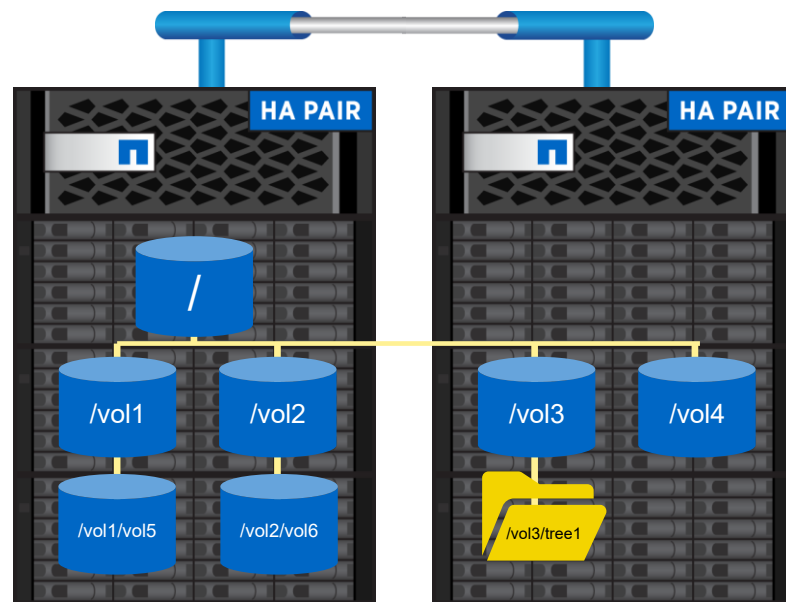
Use additional commands to show, modify, change the index, or delete the export rule.

Client match specification:

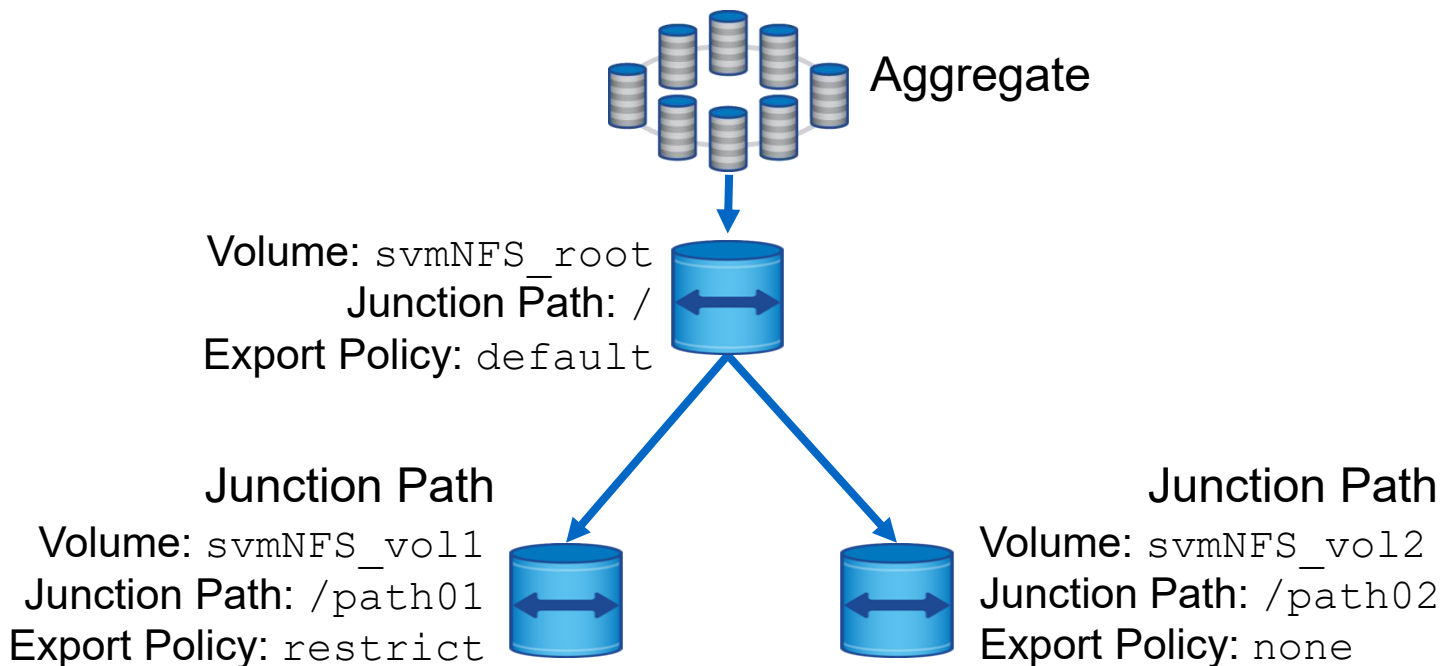
- As a hostname; for example, `host1`
- As an IPv4 address; for example, `10.1.12.24`
- As an IPv4 address with a subnet mask expressed as a number of bits; for example, `10.1.12.10/4`
- As an IPv4 address with a network mask; for example, `10.1.16.0/255.255.255.0`
- As a netgroup, with the netgroup name preceded by the `@` character; for example, `@netgroup`
- As a network name from files, Network Information System (NIS), or Lightweight Directory Access Protocol (LDAP) preceded by the `=` character; for example, `=networkname`
- As a domain name preceded by the `."` character; for example, `.example.com`

NFSv3 Exported Resource

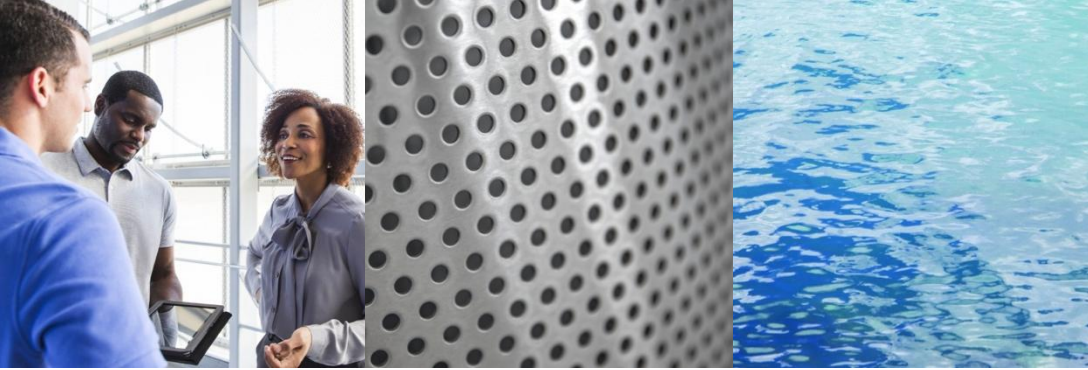
- SVM root volume
Exported root volume example: /
- Volume
Exported volume example: /vol4
- Qtree
 - Exported qtree example: /vol3/tree1
 - Supported in clustered Data ONTAP 8.3 and ONTAP 9 for NFSv3 and NFSv4



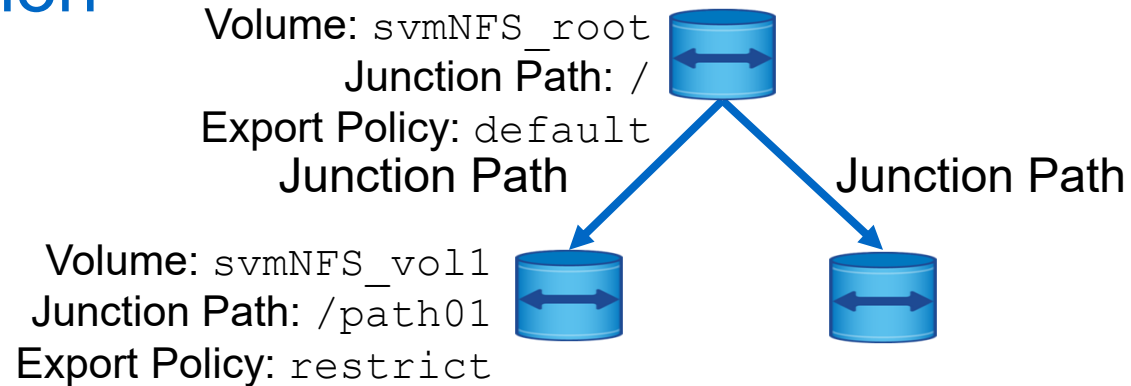
Export Policies and Volumes



The export policy can be configured at the SVM root volume level, and every subvolume and qtree in an SVM.



Discussion



What happens if you have a `default` export policy that enables all clients and a `restrict` policy that enables no client access?

Managing the Export Policy Cache

- The export policy cache stores information for faster processing.
- Cache types:
 - Access cache: Mapping of clients to corresponding export policies
 - Name: Mapping of UNIX user names to corresponding UNIX user IDs
 - ID: Mapping of UNIX user IDs to corresponding UNIX user IDs and extended UNIX group IDs
 - Host: Mapping of host names to corresponding IP addresses
 - Netgroup: Mapping of netgroups to corresponding IP addresses of members
 - Showmount: List of exported directories from the SVM namespace
- Caches have different expiration and refresh times.
- Clear the cache for an individual cache or all caches simultaneously.

```
cluster1::> vserver export-policy cache flush -vserver -node node1 -cache all
```

Mounting from a Client

To mount an export from a client:

1. Establish a session with a client.
2. Create a directory as a mount point for the storage system.
3. Mount the exported directory in the host directory that you just created.
4. Change directories to the mounted export.
5. Enter `ls -al` to verify that the storage system is mounted and accessible.

```
ssh admin@10.32.30.20
```

```
# mkdir /svmNFS-vol1-qt1
```

```
# mount -t nfs -o nfsvers=3 \
    192.168.0.161:/vol1/qtree1 /svmNFS-vol1-qt1
```

```
# cd /svmNFS-vol1-qt1
```

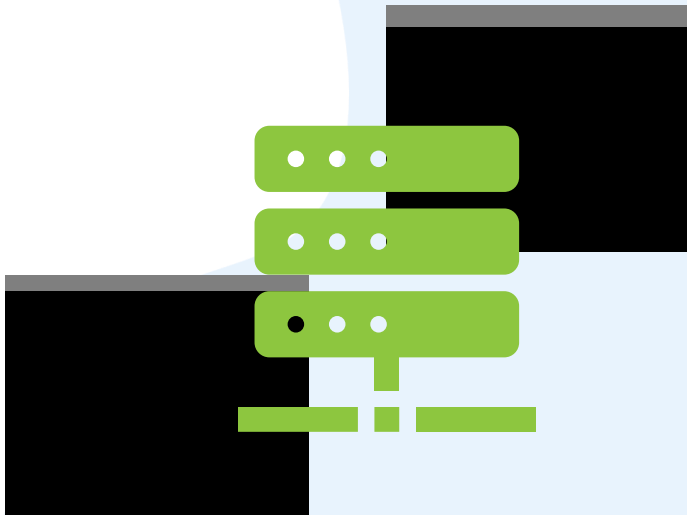
```
# ls -al
```

```
-rwxr-xr-x  root 719634  FEB 14 2012  general
-rwxr-xr-x  root 719634  FEB 14 2012  policy...
```

Red Hat Linux 6.0 and later versions default to NFSv4.

Mount Types

- Hard mounts do the following:
 - Continue to attempt to mount the file system
 - Are preferred when you mount local disks
 - Are used for reliability
 - Use `intr` to provide interrupt functionality when a hard mount stops functioning
- Soft mounts do the following:
 - Attempt to mount the file system a specific number of times and then, if they fail, report the error
 - Are preferred when you mount over NFS
 - Should not be used if frequent NFS timeouts are possible



Automatic Mounts

- `/etc/fstab`:
 - Automatically mounts the entries within the file
 - Maintains mounts; might affect boot performance
- `autofs`:
 - Automatically mounts and unmounts NFS file systems on demand, saving system resources
 - Uses these configuration steps:
 1. Install the `nfs-utils` package.
 2. Edit the `/etc/auto.master` configuration file:

```
payroll -fstype=nfs payroll:/
hr -fstype=nfs hr:/
```
 3. Start and check the status of the service:

```
# service autofs start
# service autofs status
```



Lesson 4

NFSv3 Security

Securing NFS Access with Export Policies

Create the export policy.

```
cluster1::*>vserver export-policy create ...
```

Create the export policy rule.

```
cluster1::*>vserver export-policy rule create ...  
-vserver vservername  
-policyname myexportpolicy  
-ruleindex 1  
-protocol nfs  
-clientmatch 172.16.1.0/24  
-rorule sys  
-rwrule sys  
-anon 0  
-superuser any
```

The -clientmatches

-clientmatches 1.1.1.1

-clientmatches "1.1.1.1,2.2.2.2"

-clientmatches "10.1.1.0/24,172.16.0.0/16"

-clientmatches "10.1.1.0/255.255.255.0,172.16.0.0/16"

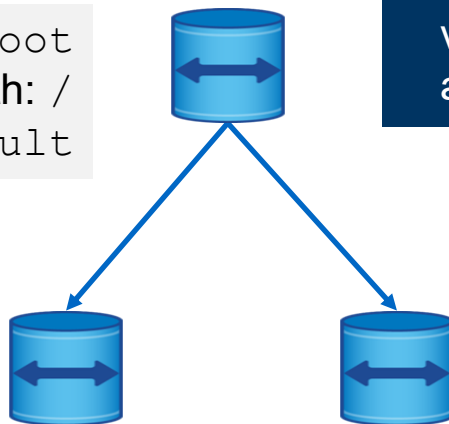
-clientmatches "@netgroup1,@netgroup2"

-clientmatches ".example.com,trainocate.com"

Namespace Export Recommendations

NetApp recommends enabling all access to the root of the SVM, by using the export rules.

Volume: `svmNFS_root`
Junction Path: `/`
Export Policy: `default`



Control read and write to the SVM root volume through UNIX permissions or access control lists (ACLs).

NetApp recommends using 0755 permissions (the default security) to restrict nonowners of the volume from writing to the SVM root volume.

User and Group Accounts

ONTAP Defaults



■ Users:

- root (0)
- pcuser (65534)
- nobody (65535)



■ Groups:

- daemon (1)
- root (0)
- pcuser (65534)
- nobody (65535)

ONTAP System Manager or the `vserver setup` command creates users and groups in ONTAP 9 software.

ONTAP Limits



■ Groups:

- AUTH_UNIX: 16
- AUTH_RPCGSS: 32

Auxiliary GIDs

Enable the auxiliary group ID (GID) option:

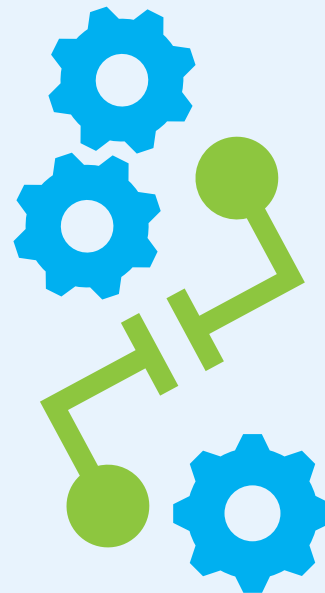
```
cluster1::*>vserver nfs modify ...  
    -is-auth-sys-extended-groups-enabled true
```

Define the limit (maximum of 1024):

```
cluster1::*>vserver nfs modify ...  
    -extended-groups-limit 32
```

Anon Setting

- Provides an effective user ID (UID) when a request arrives in one of the following states:
 - Without valid credentials
 - With a root credential (called *root squashing*)
- Is used for file and export file permissions
- Has a default value of 65534:
 - Most Linux OSs map this value to “nobody” or “nfsnobody.” ONTAP 9 software maps this value to “pcuser.”



Root User Access



User Account on Client

```
root::0:1::/:
```

- If the client root user (0) is compromised, then that user can be used to compromise security on a storage system.
- Anyone can masquerade as root (0).

User Accounts on Storage

```
root::0:1::/:
```

```
pcuser::65534:65534::/:
```

```
nobody::65535:65535::/:
```

To plug this security hole, use either of these mapping types:

- Anonymous
- Root user

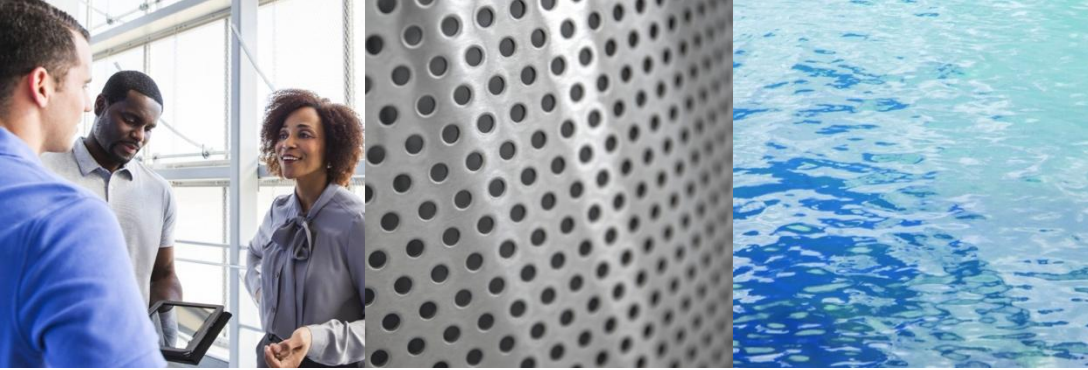
Root User Access

ONTAP

ONTAP CLI

```
cluster1::>vserver export-policy rule create...  
-superuser=any
```

Superuser Value	Root Access
any	The client root is root, regardless of security type on client (none, sys, krb5).
never	The client root is never root.
none	Squash root to anon value.
krb5 ntlm sys	Take root from the corresponding mechanism.



Take a Poll

One-Question Quiz

Duration: 15-25 minutes

Poll Questions



1. What are three features of NFSv3? (Choose three.)
 - a. It supports 64-bit file sizes and offsets.
 - b. It enables client access to files larger than 2GB.
 - c. It requires TCP.
 - d. The server can buffer multiple write requests.
 - e. It does not integrate with Lightweight Directory Access Protocol (LDAP).

Poll Questions: Answers



1. What are three features of NFSv3? (Choose three.)

- ✓ a. It supports 64-bit file sizes and offsets.
- ✓ b. It enables client access to files larger than 2GB.
- c. It requires TCP.
- ✓ d. The server can buffer multiple write requests.
- c. It does not integrate with Lightweight Directory Access Protocol (LDAP).

Poll Questions



1. What are three valid types of export policy cache? (Choose three.)
 - a. name
 - b. ID
 - c. IP
 - d. access

Poll Questions: Answers



1. What are three valid types of export policy cache? (Choose three.)

- ✓ a. name
- ✓ b. ID
- c. IP
- ✓ d. access

Poll Questions



2. Which two types of resources can be exported with ONTAP software?
(Choose two.)
- a. directory
 - b. volume
 - c. qtree
 - d. file

Poll Questions: Answers



2. Which two types of resources can be exported with ONTAP software?
(Choose two.)

- a. directory
- ✓ b. volume
- ✓ c. qtree
- d. file

Poll Questions



1. Root user access is specified in the export with which of the following command options?
 - a. anon
 - b. superuser
 - c. root
 - d. admin

Poll Questions: Answers



1. Root user access is specified in the export with which of the following command options?

a. anon

✓ b. superuser

c. root

d. admin

Module Summary



This module focused on enabling you to do the following:

- Describe the features of NFS Version 3 (NFSv3)
- Describe the implementation of NFSv3 on NetApp storage
- Describe NFS exports
- Explain NFSv3 security using ONTAP software



Complete an Exercise

Module 2: Configuring an SVM for NFSv3

Duration: 30 minutes

YOU ARE HERE

- ✓ NFS Fundamentals
- ✓ NFS Version 3
 - NFS Versions 4 and 4.1
 - Multiprotocol Configuration
 - Monitoring and Managing NFS